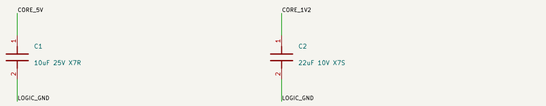
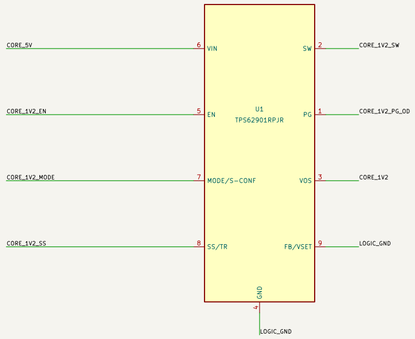
THREE LOW-VOLTAGE REGULATORS — EXTERNAL SEQUENCING REQUIRED

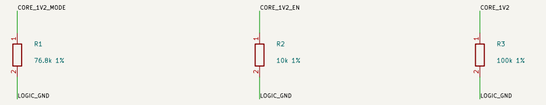
TPS62901 replaces the TPS62160 study here: tighter regulation and adjustable ramp. Frozen ledger unchanged. CORE_5V required 4.85-5.15V.

CORE_1V2: 1.2V / 0.8A reservation





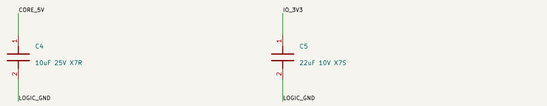




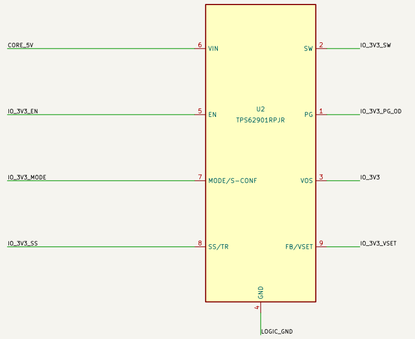


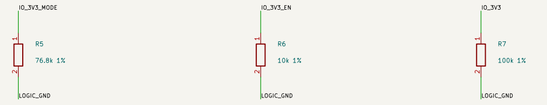
76.8k selects VSET, 1MHz forced PWM and discharge.
EN defaults off only with high-Z external controller.
PG is raw open drain: external pull-up and supervision required.

IO_3V3: 3.3V / 0.4A reservation











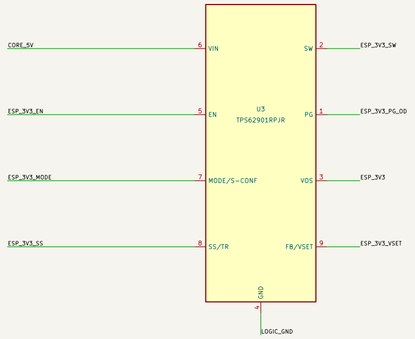


76.8k selects VSET, 1MHz forced PWM and discharge.
EN defaults off only with high-Z external controller.
PG is raw open drain: external pull-up and supervision required.

ESP_3V3: 3.3V / 0.7A reservation





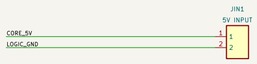




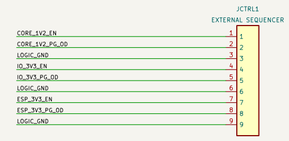


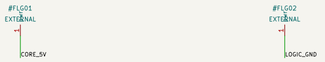


76.8k selects VSET, 1MHz forced PWM and discharge.
EN defaults off only with high-Z external controller.
PG is raw open drain: external pull-up and supervision required.









No PG-to-EN cascade. PG thresholds do not prove strict FPGA rail windows: PG is undefined below 1.8V VIN.
6.8nF soft start models 2.231ms nominal: loaded ramp, input fault and restart sequencing remain unmeasured.
VCC/PLL must precede VCCIO: FPGA reset and package supply/filter/decoupling connections are outside this block.

Discharge is conditional: requires a previous enable and input supply near 2V or above.
No guaranteed power-off discharge, latch-off, reverse-current blocking or machine permission.
ESP rail is separately regulated; independent upstream fault isolation is NOT implemented.

Engineering source; no PCB or physical qualification

Sheet: /
File: logic_power.kicad_sch

Title: Baseline logic power regulator block

Size: A2	Date: 2026-09-28	**Rev: r0.1**
KiCad E.D.A. 10.0.4		Id: 1/1